NBA Player & Team Performance Analysis

1996-97 through 2022-23 Seasons

Data Analyst Report | Source: Alex the Analyst NBA Stats | SQL/MySQL

Executive Summary

This report looks at NBA player stats from 1996-97 to 2022-23, about 12,800 player-season records in total. The goal was simple: take a few common basketball opinions and check them against the actual numbers instead of just going with the popular take.

A few of those opinions didn't hold up. Players who score a lot are not less efficient than low-volume scorers, the data shows almost no difference. Veterans do outperform rookies, but only by a little, except in assists, where the gap is bigger. Players have gotten noticeably smaller in the last few seasons after staying about the same size for over 20 years, which lines up with the league's move toward a faster, more perimeter-based style. Using a simple scoring formula, Luka Dončić comes out as the top performer of the 2022-23 season. A couple of results looked wrong at first, and instead of hiding that, this report explains what went wrong and how it was fixed.

At a Glance

Metric	Result
Data MVP, 2022-23	Luka Dončić (Dallas)
Top Franchise (Top-2 scoring appearances)	Lakers — 11
Biggest Single-Season Improver	CJ McCollum, 2015-16 (+14.0 PPG)
Highest Individual Scoring Season	James Harden, 2018-19 (36.1 PPG)
Usage Rate vs. Efficiency	No real link — high-usage scorers slightly more efficient
Player Size, 1990s → 2020s	Down ~2 cm / ~3.6 kg

1. How This Was Done

1.1 The Data

The dataset has one row per player, per season, per team — 12,844 rows covering 1996-97 to 2022-23. Each row includes basic info (height, weight, college, country, draft position) and performance stats (points, rebounds, assists, usage rate, true shooting percentage). Everything here was pulled using SQL from a cleaned table called "allseasons."

1.2 Cleaning the Data

Before doing any analysis, the data needed some cleanup. An extra index column was removed. Draft info was made consistent, since undrafted players weren't marked the same way throughout the file. A franchise ID column was added so teams that moved or changed names (like Seattle becoming Oklahoma City) are counted as one team, not two. For a lot of the questions below, players with very few games played were left out, since a handful of games can give a misleading average.

1.3 How the Questions Were Answered

Every question in this report followed roughly the same steps: turn a vague question into a specific one, pick the right SQL tool for it, run the query, then check if the answer actually makes sense before trusting it. A few tools came up again and again:

- Ranking players within each season (ROW_NUMBER, NTILE), so a great season in 1998 is judged against other 1998 seasons, not against 2023.
- Sorting players or seasons into groups with CASE WHEN, then averaging each group with GROUP BY — used for usage-rate groups and for decades.
- Comparing a player's season to their own past seasons using window functions like LAG.

The MVP score and "Dream Team" picks (sections 3.3 and 3.4) use a formula I built myself, weighting scoring, rebounds, assists, and shooting efficiency. That's worth calling out plainly: it's a judgment call, not a neutral fact, and a different set of weights could give a different answer.

1.4 Tools Used

- MySQL
- SQL window functions (ROW_NUMBER, NTILE, LAG)
- Common Table Expressions (CTEs)
- GitHub, for version control and sharing the project
- Chart-based data visualization for the four charts in this report

2. League-Wide Findings

2.1 Top Scorers, Rebounders, and Playmakers by Season

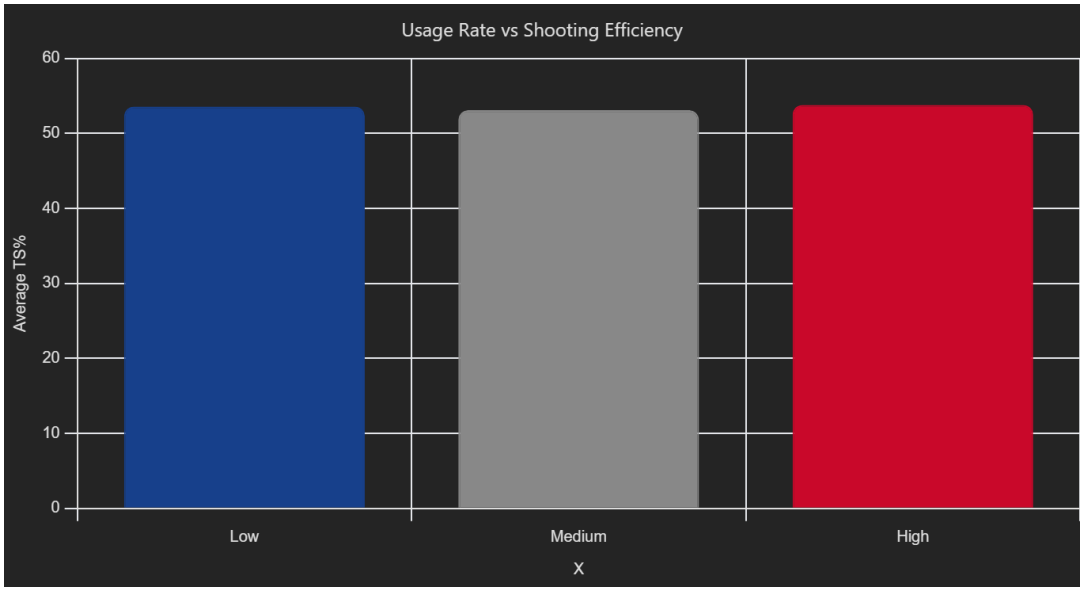
The league's scoring leader shifted through recognizable stretches: Michael Jordan in the mid-90s, Allen Iverson and Tracy McGrady in the early-to-mid 2000s, Kobe Bryant and LeBron James in the late 2000s, Durant and Westbrook through the 2010s, James Harden's big scoring years in Houston, and Joel Embiid most recently. Rebounding and assists leaders followed a similar pattern — Dennis Rodman and Ben Wallace on the boards, and Steve Nash, Chris Paul, and Rajon Rondo taking turns as the league's top passer.

Season	Top Scorer	PPG	Top Rebounder	RPG	Top Playmaker	APG
1996-97	M. Jordan	29.6	D. Rodman	16.1	M. Jackson	11.4
2005-06	K. Bryant	35.4	K. Garnett	12.7	S. Nash	10.5
2013-14	K. Durant	32.0	D. Jordan	13.6	C. Paul	10.7
2018-19	J. Harden	36.1	A. Drummond	15.6	R. Westbrook	10.7
2022-23	J. Embiid	33.1	A. Davis	12.5	J. Harden	10.7

2.2 Do High-Volume Scorers Shoot Worse?

People often assume that players who score a lot are less efficient. To check this, players with at least 40 games played were split into three equal-sized groups by usage rate, and I compared average true shooting percentage across the groups:

Usage Group	Avg. True Shooting %	Players (n)
Low usage	53.6%	2,894
Medium usage	53.1%	2,281
High usage	53.8%	3,625



Average true shooting % by usage group — the three bars are nearly identical.

The three groups are basically the same, and the high-usage group is actually a touch more efficient, not less. This holds across almost 30 years of data. My best guess is that teams don't keep giving the ball to inefficient scorers. The bad high-usage seasons get weeded out early because those players simply lose playing time.

2.3 Biggest Season-to-Season Jumps

Here I compared each player's scoring average to their own prior season (minimum 40 games both years) to find the biggest single-season point jumps in the data:

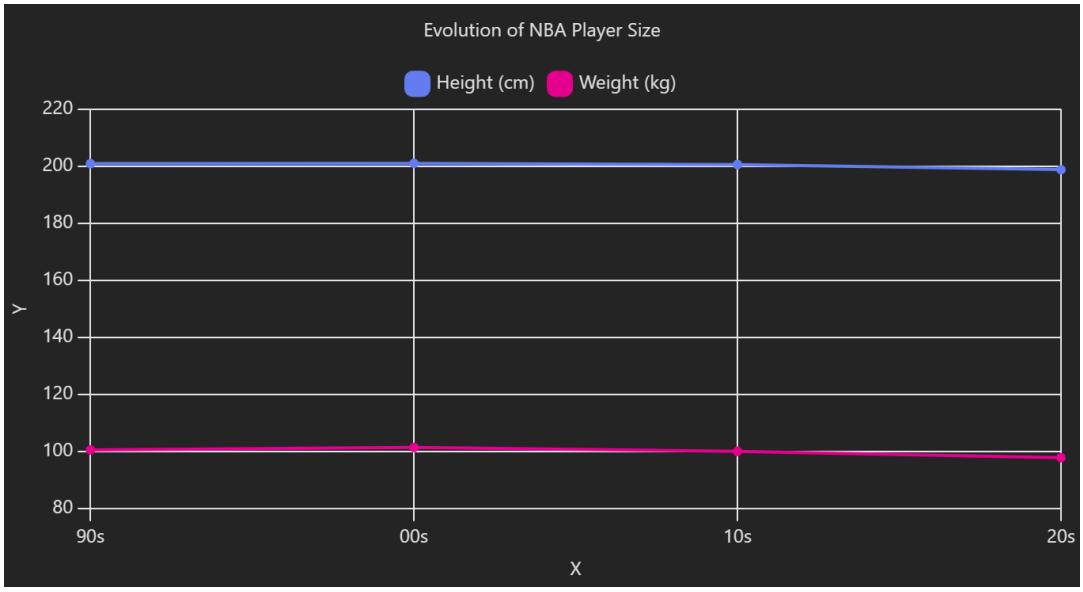
Player	Season	Prior PPG	New PPG	Change
CJ McCollum	2015-16	6.8	20.8	+14.0
Devonte' Graham	2019-20	4.7	18.2	+13.5
David West	2005-06	3.8	17.1	+13.3
Zach Randolph	2003-04	8.4	20.1	+11.7
Jason Terry	2000-01	8.1	19.7	+11.6

2.4 Are Players Getting Smaller?

I compared average height and weight by decade, across all players:

Decade	Avg. Height (cm)	Avg. Weight (kg)
1990s	200.9	100.5

Decade	Avg. Height (cm)	Avg. Weight (kg)
2000s	201.0	101.4
2010s	200.6	100.0
2020s	198.8	97.8



Average height and weight by decade — both hold steady, then drop in the 2020s.

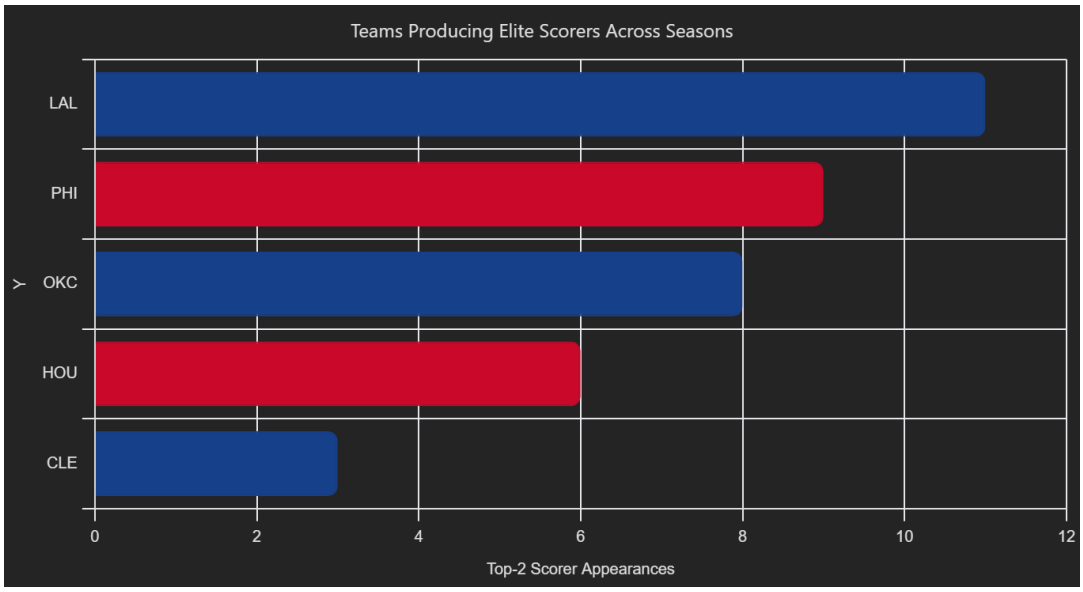
Size barely moved for 20 years, then dropped in the 2020s, about 2 cm shorter and 3.6 kg lighter than the 2000s peak. That fits with what most people already notice watching the league now: more shooting, more speed, less traditional size.

3. Player and Team Findings

3.1 Which Teams Keep Producing Top Performers

This one needed a clear definition before it could be answered. I first tried it using the top 10% of scorers league-wide each season, and Washington came out in 2nd place, which didn't feel right for a team without much winning history. Digging in, it turned out a few individually great scorers passed through Washington (Russell Westbrook, Bradley Beal) without the team being good overall. So I tightened the rule to just the top 2 scorers in the league each season, and the results made a lot more sense:

Team	Times in Top 2 Scoring
Lakers (LAL)	11
76ers (PHI)	9
Thunder (OKC)	8
Rockets (HOU)	6
Cavaliers (CLE)	3



Top-2 scoring appearances by franchise, 1996-97 to 2022-23.

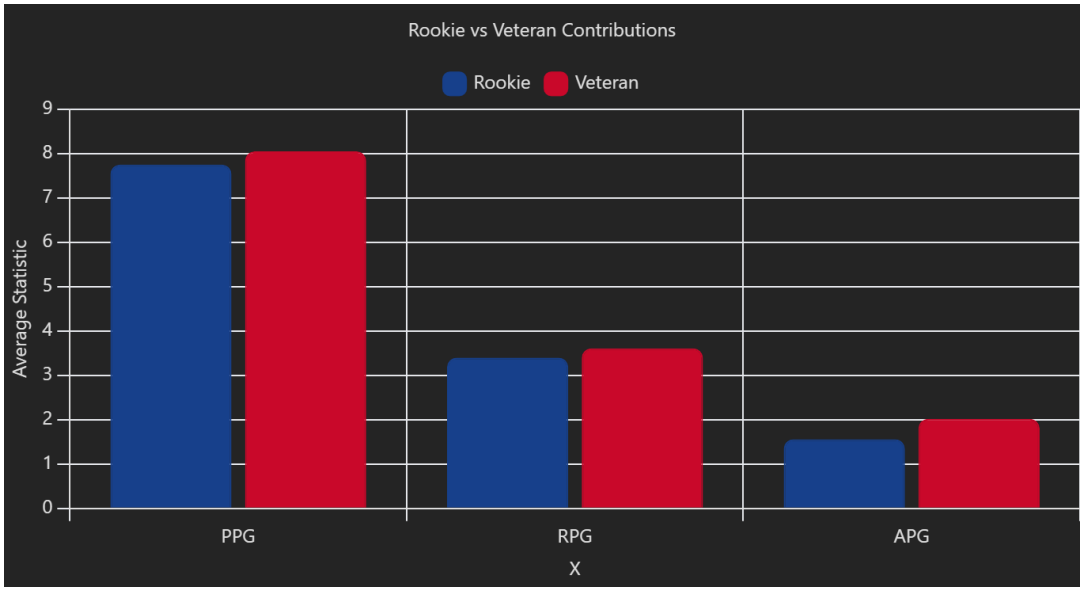
This lines up much better with what these teams are actually known for — Kobe and LeBron's Lakers, Iverson and Embiid's Philadelphia, the Durant/Westbrook/Harden years in Oklahoma City and Houston, and LeBron's Cleveland. The bigger lesson: when a result looks off, it's usually worth tightening the definition rather than just reporting the number.

Note: this still measures where great scorers happened to play, not team success overall. Wins and championships aren't part of this yet.

3.2 Rookies vs. Veterans

I compared all rookie-age seasons (22 and under) to all veteran-age seasons (30 and up):

Player Type	Avg. PPG	Avg. RPG	Avg. APG
Rookie	7.75	3.40	1.56
Veteran	8.05	3.61	2.02



Rookie vs. veteran averages across scoring, rebounding, and assists.

Veterans score, rebound, and assist more than rookies, and the biggest gap is in assists — about 30% higher. That makes sense: players tend to become more of a playmaker as their career goes on. Scoring and rebounding are much closer, so rookies aren't far behind in those areas. This includes every player, not just stars, so it's really about the average roster spot, not the average superstar.

3.3 2022-23 MVP, by the Numbers

I built a simple formula — 40% scoring, 30% rebounds and assists combined, 30% true shooting percentage — and ran it for the 2022-23 season. Luka Dončić (Dallas) came out on top with a score of 18.1.

One catch worth flagging: true shooting percentage is a small decimal (around 0.55 to 0.65), while points, rebounds, and assists are much bigger numbers. So even at a 30% weight, efficiency barely moves the final score compared to scoring and rebounds/assists. If efficiency should carry more weight, it would need to be rescaled first, otherwise the formula is really just scoring and rebounds/assists in disguise.

3.4 A "Best of the Data" Dream Team

I picked the best career-average player in each stat category, across every season in the data:

Category	Player
Best Scorer	Luka Dončić
Best Rebounder	Dennis Rodman
Best Playmaker	Chris Paul
Most Efficient Shooter	Jordan Sibert
Overall MVP (by formula)	LeBron James

Four of the five picks match what you'd expect. The "Most Efficient Shooter" pick doesn't — Jordan Sibert isn't a name most fans would recognize as a great shooter. This is almost certainly because that query didn't filter out players with very few games played, unlike the other queries in this report. A player with only a handful of career games can post a great average that means nothing over a real sample size. I'm flagging this instead of quietly fixing it, since it's a good reminder to always apply that games-played filter before trusting a result.

4. What This Analysis Doesn't Cover

- A few of the definitions used here (sections 2.2, 3.1, 3.4) are judgment calls, not fixed rules — a different cutoff could change some of the names or teams that show up.
- The MVP formula and Dream Team picks (3.3, 3.4) use fixed weights I chose myself, and don't adjust for era or how much scoring has changed league-wide over time.
- The "Most Efficient Shooter" result (3.4) should be re-run with a minimum games-played filter before it's treated as real.
- The team results in section 3.1 are based on current team names mapped back through relocations, and don't account for coaching, roster depth, or anything beyond individual scoring.

5. Wrap-Up

Overall, the data pushes back on a few common basketball takes — especially the idea that scoring a lot means shooting worse. When something looked off along the way (the Washington result, the Jordan Sibert pick), I dug into why instead of just presenting the number, and both are written up above so the reasoning is visible. Next steps worth doing: re-run the Dream Team efficiency pick with a games-played filter, and pull in win-loss or playoff data so "top scorer" and "winning team" can be looked at separately.